

ABSTRACT

The Bluetooth-Controlled Smart Navigation Robot Using ESP32 is an intelligent robotic system designed to provide wireless control and automated navigation using Bluetooth communication technology. The project integrates an ESP32 microcontroller, L298N motor driver, BO motors, a 4WD robotic chassis, lithium battery power supply, and a mobile application developed using MIT App Inventor. The primary objective of the project is to develop a cost-effective and user-friendly robotic vehicle that can be controlled remotely through a smartphone.

The robot receives commands from the mobile application through Bluetooth communication and performs various movements such as forward, backward, left, right, and stop. In addition to manual control, the system supports predefined destination-based navigation, allowing the robot to move toward locations such as Home, School, Office, and Hospital based on programmed instructions. The ESP32 processes the received commands and controls the motor driver to achieve accurate movement and navigation.

This project provides practical knowledge in embedded systems, robotics, wireless communication, and mobile application development. It demonstrates the integration of hardware and software components to create a functional robotic platform suitable for educational and research applications. Future enhancements may include Internet of Things (IoT) integration, GPS-based navigation, obstacle detection and avoidance, voice control, camera modules, and fully autonomous operation, making the system more intelligent and efficient for real-world applications.